

Eigenvectors and Linear Transformations

The Matrix of a Linear Transformation

$$[T(\mathbf{x})]_{\mathcal{B}} = M[\mathbf{x}]_{\mathcal{B}}$$

where

$$M = [[T(\mathbf{b}_1)]_{\mathcal{B}} \ [T(\mathbf{b}_2)]_{\mathcal{B}} \ \cdots \ [T(\mathbf{b}_n)]_{\mathcal{B}}]$$

The matrix M is a matrix representation of T , called the **matrix for T relative to the basis $\mathcal{B}$** and denoted by $[T]_{\mathcal{B}}$. See Figure 2.

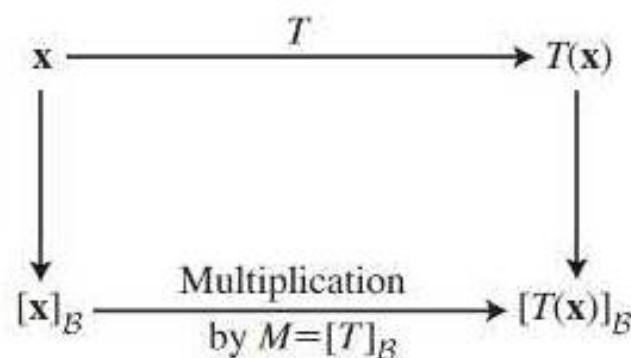


FIGURE 2

EXAMPLE 2 Suppose $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ is a basis for V . Let $T : V \rightarrow V$ be a linear transformation with the property that

$$T(\mathbf{b}_1) = 3\mathbf{b}_1 - 2\mathbf{b}_2 \quad \text{and} \quad T(\mathbf{b}_2) = 4\mathbf{b}_1 + 7\mathbf{b}_2$$

Find the matrix M for T relative to $\mathcal{B}$.

SOLUTION The $\mathcal{B}$ -coordinate vectors of the *images* of $\mathbf{b}_1$ and $\mathbf{b}_2$ are

$$[T(\mathbf{b}_1)]_{\mathcal{B}} = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \quad \text{and} \quad [T(\mathbf{b}_2)]_{\mathcal{B}} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

Hence

$$M = \begin{bmatrix} 3 & 4 \\ -2 & 7 \end{bmatrix}$$



EXAMPLE 3 The mapping $T : \mathbb{P}_2 \rightarrow \mathbb{P}_2$ defined by

$$T(a_0 + a_1t + a_2t^2) = a_1 + 2a_2t$$

is a linear transformation. (Calculus students will recognize T as the differentiation operator.)

- Find the $\mathcal{B}$ -matrix for T , when $\mathcal{B}$ is the basis $\{1, t, t^2\}$.
- Verify that $[T(\mathbf{p})]_{\mathcal{B}} = [T]_{\mathcal{B}}[\mathbf{p}]_{\mathcal{B}}$ for each $\mathbf{p}$ in $\mathbb{P}_2$.

SOLUTION

- Compute the images of the basis vectors:

$$\begin{aligned} T(1) &= 0 && \text{The zero polynomial} \\ T(t) &= 1 && \text{The polynomial whose value is always 1} \\ T(t^2) &= 2t \end{aligned}$$

$$[T(1)]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, [T(t)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, [T(t^2)]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$
$$[T]_{\mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Apply
transformation on
vector basis

Rewrite the images
as coordinates

coordinates of polynomial coefficients = $\begin{bmatrix} \text{the constant term} \\ \text{coefficient of } t \\ \text{coefficient of } t^2 \\ \vdots \\ \text{coefficient of } t^n \end{bmatrix}$

- For a general $\mathbf{p}(t) = a_0 + a_1t + a_2t^2$

$$\begin{aligned} [T(\mathbf{p})]_{\mathcal{B}} &= [a_1 + 2a_2t]_{\mathcal{B}} = \begin{bmatrix} a_1 \\ 2a_2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = [T]_{\mathcal{B}}[\mathbf{p}]_{\mathcal{B}} \end{aligned}$$

In the previous examples, the transformation was given as description, **but the standard matrix A was NOT given.** Therefore, the way to find the matrix M (the B-Matrix – the matrix of T relative to basis) is by applying the following steps:

1. Find the image of the vectors of the basis.
2. Find the coordinates of those images. They are the columns of M .

However, in the next examples, the matrix A is already given. Therefore, we can find the matrix M as a similar matrix to A as follows:

$$M = P^{-1} A P$$

OR

$$A = P M P^{-1}$$

THEOREM 8

Diagonal Matrix Representation

Suppose $A = PDP^{-1}$, where D is a diagonal $n \times n$ matrix. If $\mathcal{B}$ is the basis for $\mathbb{R}^n$ formed from the columns of P , then D is the $\mathcal{B}$ -matrix for the transformation $\mathbf{x} \mapsto A\mathbf{x}$.

EXAMPLE 4 Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Find a basis $\mathcal{B}$ for $\mathbb{R}^2$ with the property that the $\mathcal{B}$ -matrix for T is a diagonal matrix.

This question is equivalent to **asking us to diagonalize A** because when we diagonalize A , we get two matrices: P , which is a basis of $\mathbb{R}^n$ (Theorem 7c), and D , which is considered as the $\mathcal{B}$ -matrix of the transformation (Theorem 8). And both matrices are requested in this question

Step 1. Find the eigenvalues of A .

$$\det(A - \lambda I) = \det \begin{bmatrix} 7 - \lambda & 2 \\ -4 & 1 - \lambda \end{bmatrix} = 0$$

$$(7 - \lambda)(1 - \lambda) - (-8) = 0$$

$$\lambda^2 - 8\lambda + 15 = 0$$

$$(\lambda - 5)(\lambda - 3) = 0$$

the eigenvalues are: $\lambda = 5$ and $\lambda = 3$

Step 2. Find three linearly independent eigenvectors of A .

Solve equation $(A - (5)I)\mathbf{x} = 0$
Write the solution in parametric vector form. Then, take the vectors as basis:

```
>> rref([ [ 7 2; -4 1 ] - 5*eye(2) [0;0]])
ans =
     1     1     0
     0     0     0
```

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

choose $x_2 = -1$, $\therefore \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Solve equation $(A - (3)I)\mathbf{x} = 0$
Write the solution in parametric vector form. Then, take the vectors as basis:

```
>> rref([ [ 7 2; -4 1 ] - 3*eye(2) [0;0]])
ans =
     1.0000     0.5000     0
     0         0         0
```

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -0.5x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -0.5 \\ 1 \end{bmatrix}$$

choose $x_2 = -2$, $\therefore \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

Why do we choose these values for x_2 ?
Because you can choose any nonzero value

Step 3. Construct P from the vectors in step 2.

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\text{basis } \mathcal{B} \text{ for } \mathbb{R}^2 = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right\}$$

$\mathcal{B}$ -matrix for T is a diagonal matrix.

Similarity of Matrix Representations

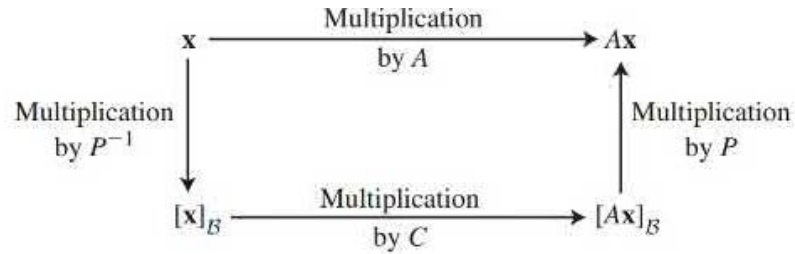


FIGURE 4 Similarity of two matrix representations:
 $A = PCP^{-1}$.

in this example, the matrix A **is already given**.
Therefore, we can find the matrix M as follows:

$$M = P^{-1}AP$$

The set of all matrices similar to a matrix A coincides with the set of all matrix representations of the transformation $\mathbf{x} \mapsto A\mathbf{x}$.

EXAMPLE 5 Let $A = \begin{bmatrix} 4 & -9 \\ 4 & -8 \end{bmatrix}$, $\mathbf{b}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, and $\mathbf{b}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. The characteristic polynomial of A is $(\lambda + 2)^2$, but the eigenspace for the eigenvalue -2 is only one-dimensional; so A is not diagonalizable. However, the basis $B = \{\mathbf{b}_1, \mathbf{b}_2\}$ has the property that the B -matrix for the transformation $\mathbf{x} \mapsto A\mathbf{x}$ is a triangular matrix called the *Jordan form* of A .¹ Find this B -matrix.

SOLUTION If $P = [\mathbf{b}_1 \quad \mathbf{b}_2]$, then the B -matrix is $P^{-1}AP$. Compute

$$AP = \begin{bmatrix} 4 & -9 \\ 4 & -8 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -6 & -1 \\ -4 & 0 \end{bmatrix}$$
$$P^{-1}AP = \begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} -6 & -1 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}$$

Notice that the eigenvalue of A is on the diagonal. ■